



Introduction to Computer Science

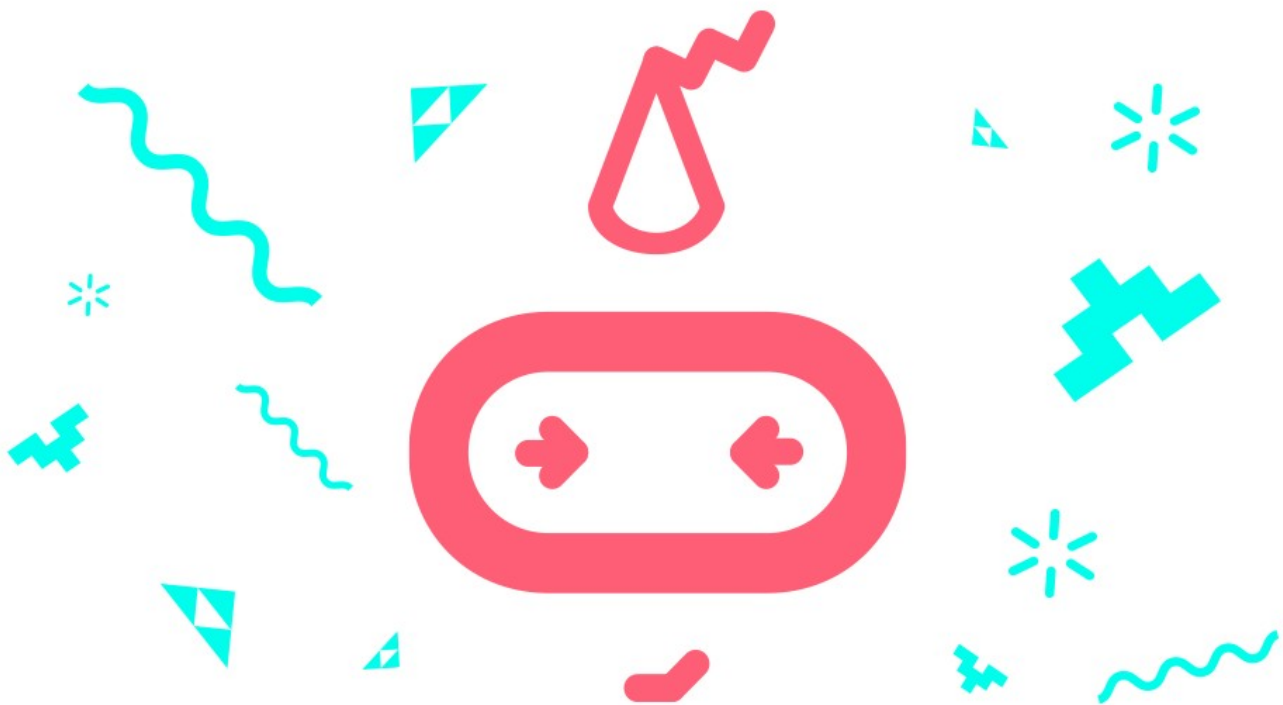
Unit 12: Accelerometer

Student workbook
makecode.microbit.org



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Overview

Unit summary

This unit introduces the accelerometer functionality of the micro:bit. Even if you're unfamiliar with the word accelerometer, you may be aware of the function. The micro:bit accelerometer measures how the micro:bit is positioned and moving through space. In the unplugged activity, you will sense your own body's way of knowing its position and movement through space. The birdhouse activity will teach you to use the micro:bit's accelerometer capabilities to create a program that reminds you to stand up every so often. This unit's project is to create a "multi-tool" that uses a combination of different sensors, including the accelerometer, to solve a problem or serve a purpose.

Learning goals

During this unit, you will:

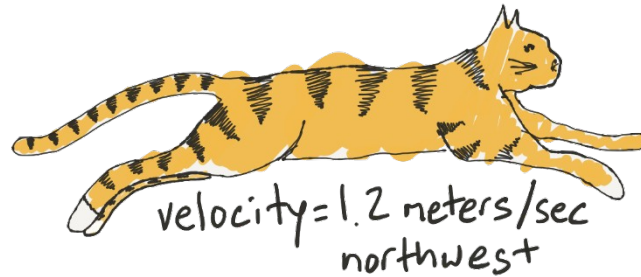
- Understand how to use the Accelerometer blocks to sense the micro:bit's position and movement in three-dimensional space
- Understand the x, y, and z axes and measurement of gravitational force
- Apply the above knowledge and skills to design a unique program using the accelerometer

Lesson A: Understanding the accelerometer

Overview: Accelerometer

An accelerometer is a device that measures *acceleration*. Acceleration is itself a measure of how an object's velocity changes.

Velocity describes an object's current speed in a particular direction. For example, you could describe the velocity of a car as "headed north at 50 mph".



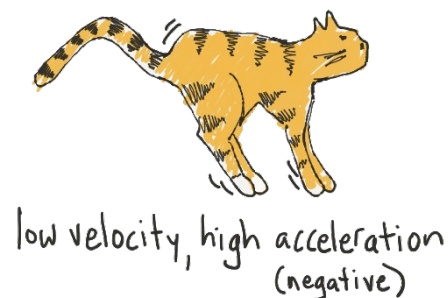
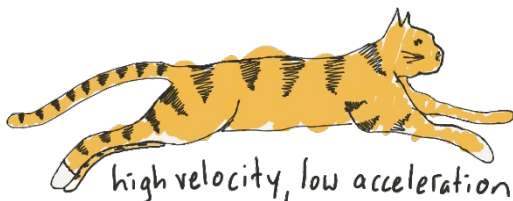
Acceleration is a measure of the rate at which the car's speed and/or direction changes.

In all these cases an acceleration occurs:

- Direction change: The car speed continues at 50 mph, yet the car turns to travel in a northeast direction
- Speed change: The car speed changes to 45 mph and the car continues heading north
- Speed and direction change: The car speed changes to 65 mph and the car turns to travel in a northeast direction

Acceleration is a measure of the rate of these changes over time.

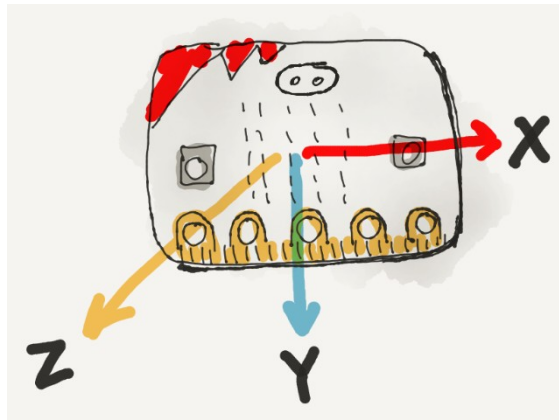
Note: Whether the car speeds up or slows down, an acceleration occurs. Students may think acceleration is only happening when something speeds up!



You measure acceleration with the milli-g, which is 1/1000 of a g. A g is as much acceleration as you get from Earth's gravity.

The micro:bit measures the acceleration value (milli g-force) in one of three dimensions or the combined force in all directions (x, y, and z).

When the micro:bit is flat on a table with the screen pointing up, the gravity force is aligned with the Z axis of the micro:bit.



If you tilt it up and down, the force will align with the Y axis—this is how we can detect tilting! As the force along Y grows, the micro:bit is tilting more and more vertically.

Parameters

- **dimension:** The direction you are checking for acceleration or the total strength of force
- **x:** Acceleration in the left and right direction
- **y:** Acceleration in the forward and backward direction
- **z:** Acceleration in the up and down direction
- **strength:** The resulting strength of acceleration from all three dimensions (directions)

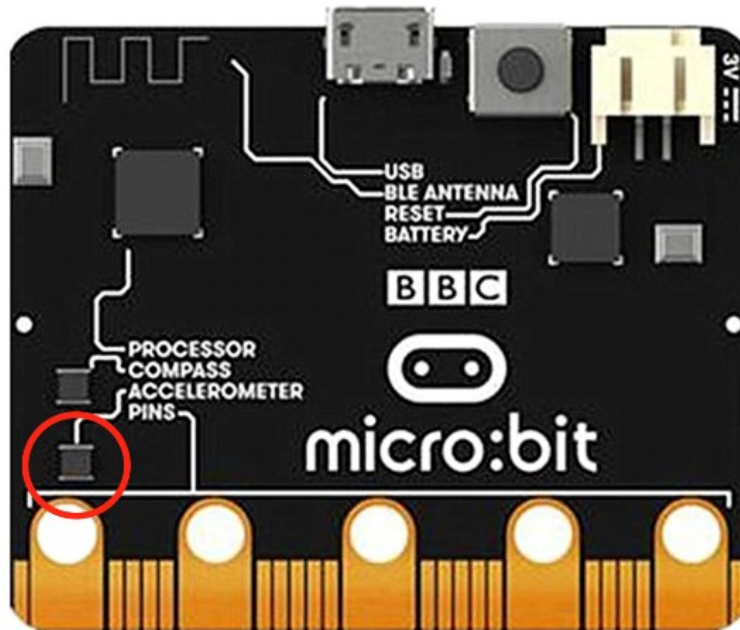
The micro:bit's accelerometer also measures how fast the micro:bit is speeding up or slowing down as it moves through space.

Output

The accelerometer feature returns a number that represents the amount of acceleration. For example: When the micro:bit is lying flat on a surface with the screen pointing up:

- **x** is 0
- **y** is 0
- **z** is -1023
- **strength** is 1023

On the back of the micro:bit, look closely in the lower left corner for the accelerometer.



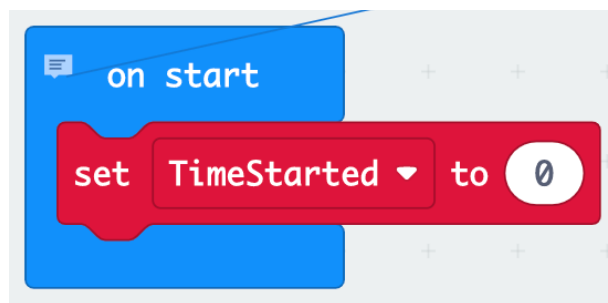
Lesson B: Explore the accelerometer tools

Coding activity: Stand for Health!

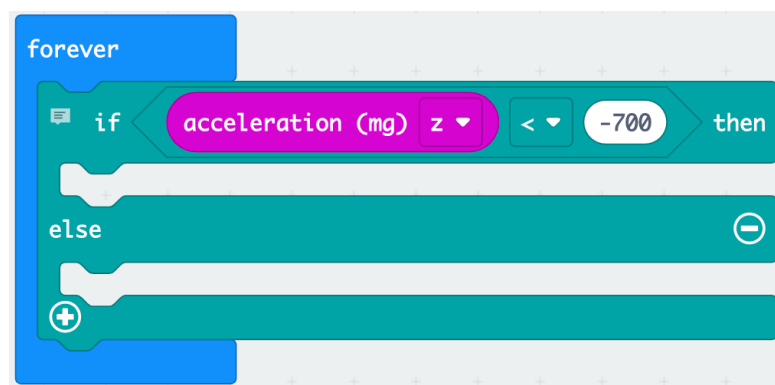
In our modern world, most of us sit for long stretches of time without ever getting up to stretch our legs. Standing up every so often is good for our physical health.

We will use the micro:bit's accelerometer to create a program that will let us know if we have been sitting too long.

1. Inside the Variables Toolbox: Create a variable to keep track of how much time has passed since the program started.
 - a. Name the variable TimeStarted.
 - b. Place the newly created 'set TimeStarted' block inside the 'on start' block.
 - c. Set the value of this variable to 0 (zero) each time the program starts.



2. From the Logic toolbox: Get and place an 'if...else' block inside the 'forever' block.
 - a. From the Logic toolbox: Get and place a comparison block into the 'true' space in the 'if' line of code.
 - b. From the Input Toolbox: Get and place the acceleration block into the left side of the comparison block.
 - c. In the acceleration block: Change the default value of 'x' to 'z'.
 - d. Set the comparison block symbol to 'less than' (<).
 - e. Set the value on the right-hand side of the comparison block to -700.



Check for understanding:

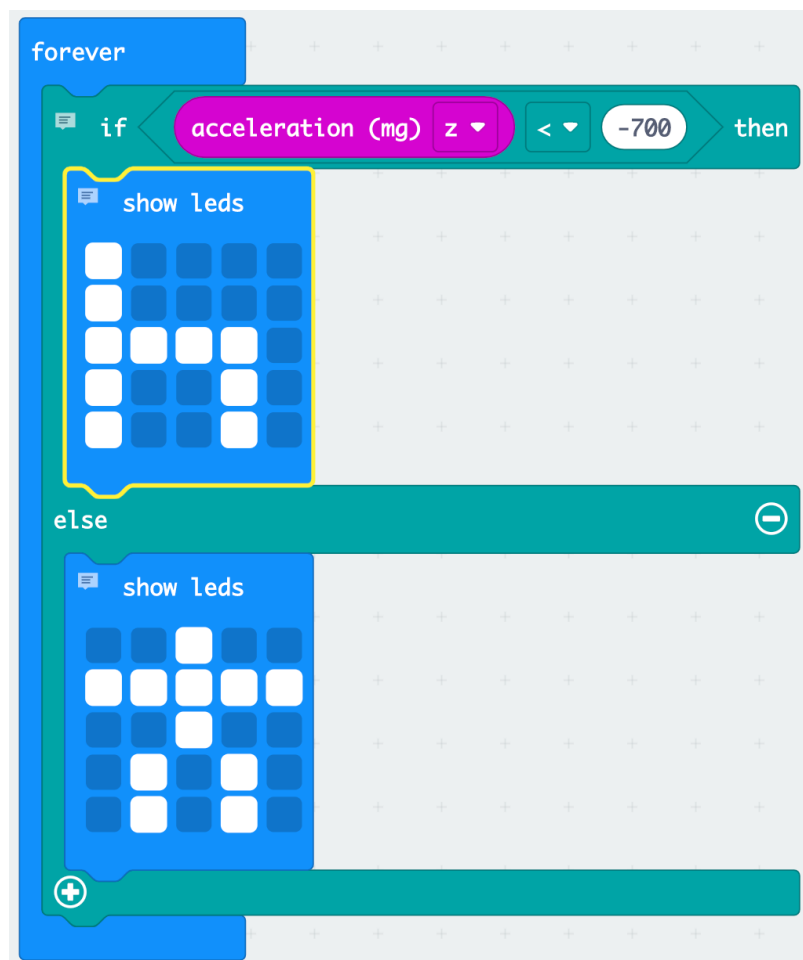
This line of code tells the micro:bit to forever check the acceleration of the micro:bit and if the value of acceleration is less than -700, do something.

Remember that when the micro:bit is lying flat on a surface with the screen pointing up:

x is 0, **y** is 0, **z** is -1023, and **strength** is 1023.

If the micro:bit is lying face up and flat, the Z value will be in this negative range. When your program is done, you can experiment with this number value, making it greater or less than the value used here to see how changing the value affects how the micro:bit reacts.

3. From the Basic Toolbox: Get two 'show leds' blocks.
 - a. Place one 'show leds' block under the 'if' line of code.
 - b. Make an image of a chair in this block to indicate sitting. (Feel free to change it to an image that works for you.)
 - c. Place the other 'show leds' block under the 'else' line of code.
 - d. Make an image of a person standing in this block. (Feel free to change it to an image that works for you.)

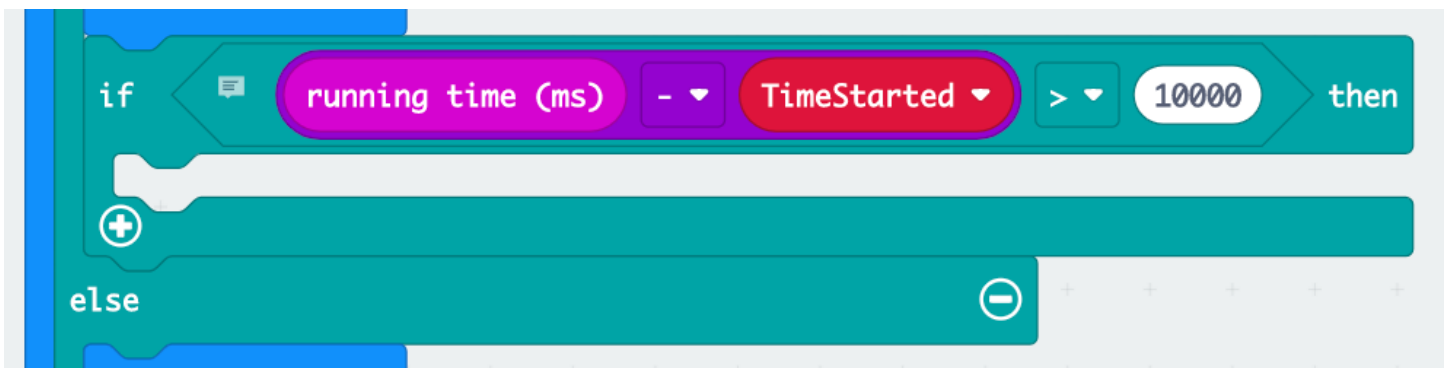


Check for understanding:

Now our micro:bit will show the chair image if the micro:bit is lying face up and flat, and a person standing image if the position of the micro:bit changes from lying flat and face up.

Now, we will add in the time variable to keep track of how long we have been sitting.

4. From the Logic toolbox: Get and place an 'if' block directly under the first 'show leds' block.
 - a. From the Logic toolbox: Get and place a comparison block into the 'true' space in this 'if' line of code.
 - b. From the Math Toolbox: Get and place an operation block into the left side of the comparison block.
 - c. From the Input...more Toolbox: Get and place a 'running time(ms)' block into the left side of the math operation block.
 - d. Set the math operation to minus (-).
 - e. From the Variables Toolbox: Get and place a 'TimeStarted' block into the right side of the math operations block.
 - f. Set the comparison block symbol to 'greater than' (>).
 - g. Set the value on the right-hand side of the comparison block to 10000 (ten seconds).



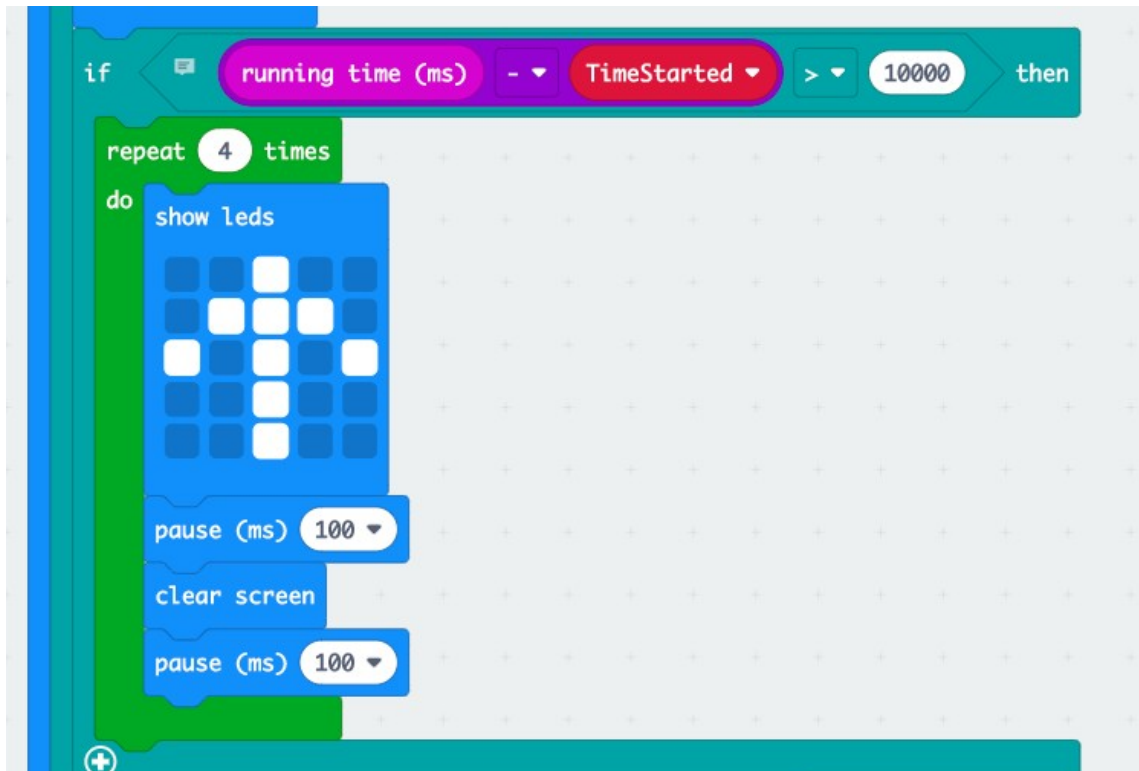
Check for understanding:

This if statement tells the micro:bit to check the difference between when we started the program and now. If the difference between when the program started and now is greater than 10 seconds, do something.

We will have the micro:bit flash a message to indicate that you have been sitting longer than the desired time.

NOTE: We're using 10 seconds here for demonstration and testing purposes. In reality, you would want to set the time somewhere closer to half an hour, which is 1,800,000 milliseconds.

5. From the Loops Toolbox: Get and place a 'repeat' block under the if statement we just created.
 - a. From the Basic Toolbox: Get one 'show leds' block, one 'clear screen' block, and two 'pause' blocks.
 - b. Place the 'show leds' block inside the 'repeat' block.
 - c. Create an arrow image or an image of your choosing in this block. This image will flash on the micro:bit screen to alert the user that it's time to stand up!
 - d. Place one of the 'pause' blocks below this 'show leds' block.
 - e. Place the 'clear screen' block just below the 'pause' block.
 - f. Place the second 'pause' block just below the 'clear screen' block.

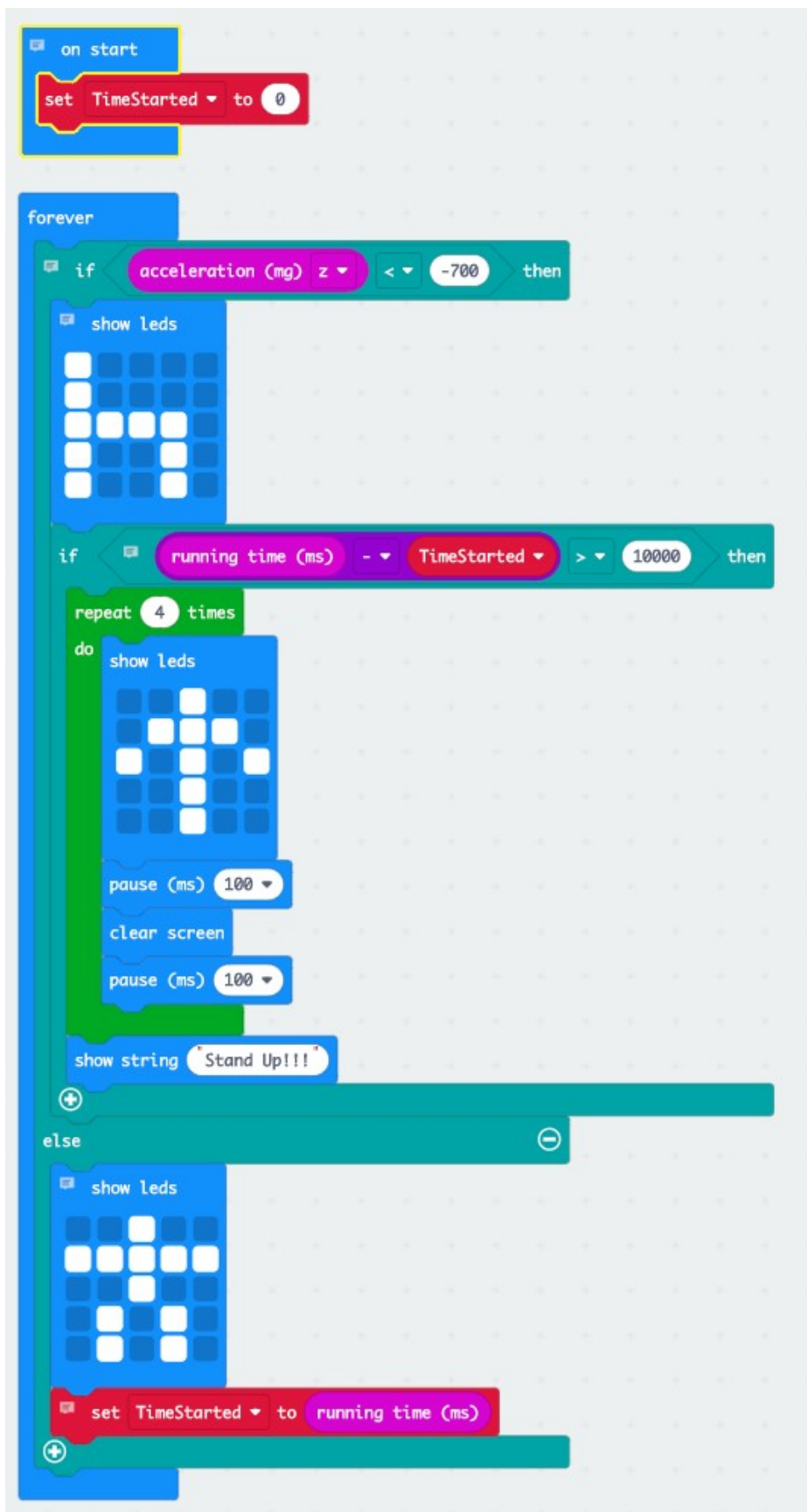


In addition to this flashing image, let's add a message!

6. From the Basic Toolbox: Get and place a 'show string' block below the 'repeat' loop block.
 - a. Write a short message reminding the user to stand up!
 - b. When the user stands up, they will see the person standing image.
 - c. In case the user sits back down again, we need to reset the TimeStarted variable to the current program runtime.
 - d. From the Variables Toolbox: Get and place a 'set TimeStarted' block just below the image of the person standing in the else section of the code.
 - e. From the Input...more Toolbox: Get and place a 'running time(ms)' block into the right side of the 'set TimeStarted' block.

Note: You can also manually restart the program each time you sit back down by pressing the micro:bit's reset button or by turning the battery pack on and off (if your battery pack has an on/off switch.)

Complete program



Solution link: makecode.microbit.org/_5gs2WR1fM8uy

Test the code

Run the code in the Simulator first. Then download the program to the micro:bits and test.

Try it!

1. Tape the micro:bit (and battery pack) to the user's leg.
2. Start the program.
3. After ten seconds, the micro:bit image should start flashing and your message should appear.
4. Stand up!
5. The standing person image should appear.

Complete code link: https://makecode.microbit.org/_XMf5wFFD01pU

Lesson C: Make an accelerometer project

Overview: Global Goals

THE GLOBAL GOALS For Sustainable Development



In 2015, world leaders got together and came to agreement on 17 major global issues that need to be addressed. Visit the Global Goals website: www.globalgoals.org

- Each of the 17 goals (globalgoals.org/goals) is broken down into smaller, more specific goals.
- The webpages for each of these more specific goals describe ways that citizens can take action to help reach that goal.

After exploring the website, consider the following questions:

- Which of the 17 goals interest you the most?
- Which of the 17 goals do you feel most directly affect you and your community?
- Why do you think they refer to these global issues as “goals to reach” rather than “problems to solve”? How does describing something as a “goal” differ from calling it a “problem”?

Global Goals and the micro:bit: do your :bit!

As we have experienced already, the micro:bit is for making. From tools to games, you have made different projects and products using the micro:bit. The micro:bit can also be used to make a difference!

do your :bit



You can use your micro:bit to help reach one of the 17 Global Goals!

Visit the "do your :bit" website (microbit.org/projects/do-your-bit) to find out how other students from around the world have created projects that aim to help reach these goals.

Explore the videos, resources, challenges, and projects to get ideas for a project of your own.

Activity: micro:bit accelerometer project

For this project, you will work separately or in pairs to design a project that incorporates the micro:bit's accelerometer capabilities.

Project expectations

Follow the design thinking approach and make sure your project meets these specifications:

- Uses the accelerometer to gather input about the micro:bit's position and motion in space
- Uses accelerometer blocks in a way that is integral to the program
- The program compiles and runs as intended and includes meaningful comments in the code
- Provide the written Reflection Diary entry (which we'll talk about after you complete your project)

The design thinking process:

1. **Empathize** by learning more about your target audience
2. **Define**: Understand and identify your audience's problems or needs
3. **Ideate**: Brainstorm several possible creative solutions
4. **Prototype**: Construct rough drafts or sketches of your ideas
5. **Test** your prototype solutions

Use this space to track your design thinking process:

Project examples

Complete programs

Following are two complete project ideas.

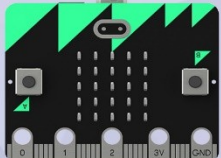
Both can be found in the Coding Cards section on the home page of the MakeCode for micro:bit site:
makecode.microbit.org

MAKING MICRO:BIT GAMES



SHAKE THE BOTTLE

This game is also inspired by a game in the Nintendo Switch game 12Switch. Players take it in turn to hold the micro:bit, which could be taped to a bottle. They give it a shake and some bubbles show on the screen. They pass it to the next player who also gives it a shake. At some point the bottle will pop and the player left holding the bottle loses.

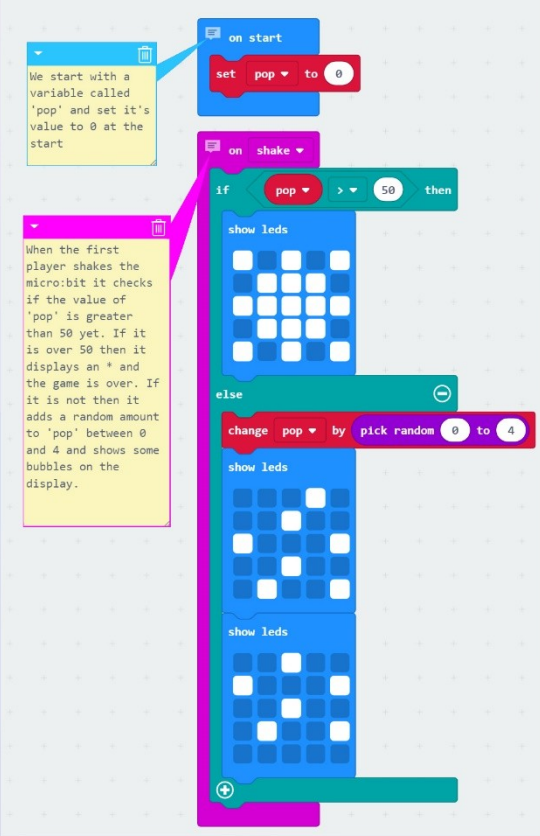


There are no extra components needed for this game as it is just using the built in accelerometer to detect when the bottle is shaken.

@stulowe80

We start with a variable called 'pop' and set it's value to 0 at the start

When the first player shakes the micro:bit it checks if the value of 'pop' is greater than 50 yet. If it is over 50 then it displays an * and the game is over. If it is not then it adds a random amount to 'pop' between 0 and 4 and shows some bubbles on the display.



This game is inspired by a game in the Nintendo Switch game 12Switch. Players take it in turns to hold the micro:bit, which could be taped to a bottle. They give it a shake and some bubbles show on the screen. They pass it to the next player, who also gives it a shake. At some point, the bottle will pop and the player left holding the bottle loses.

There are no extra components needed for this game as it just uses the built-in accelerometer to detect when the bottle is shaken.

The code

We start with a variable called 'pop' and set its value to 0 at the start. When the first player shakes the micro:bit, it checks if the value of 'pop' is greater than 50. If it is over 50, then it displays an asterisk (*) and the game is over. If it is not over 50, it adds a random amount to 'pop' between 0 and 4 and shows some bubbles on display.

Credit: Twitter user @stulowe80

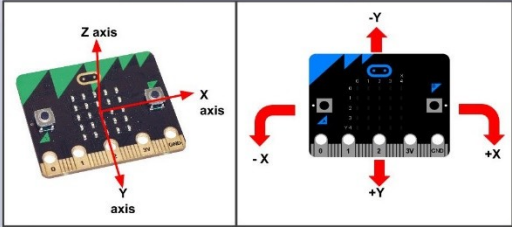
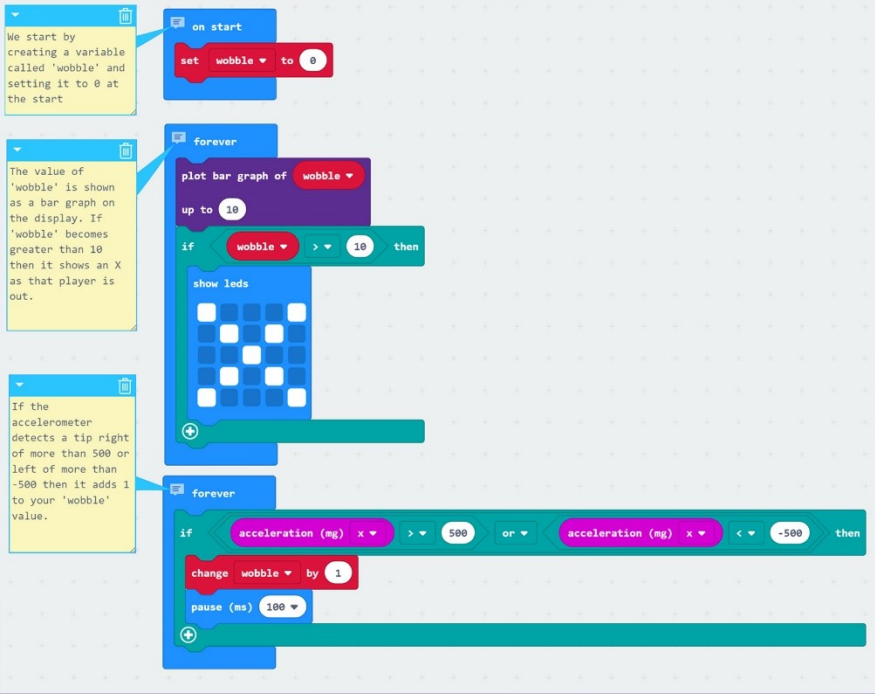
MAKING MICRO:BIT GAMES

ZEN

micro:bit @BHS

ZEN is a game our students made that was inspired by the Nintendo Switch game 12Switch. The aim of the game is to strike a yoga pose and hold it while keeping the Micro:bit completely still. If you wobble your screen starts to fill up. If it is full then you are out.

There are no extra components needed for this game as it is just using the built in accelerometer on the x axis to detect how much it wobbles rotationally left and right. You could tape the micro:bit and battery to a piece of card to make it easier to hold flat.



@stulowe80

ZEN is a game our students made that was inspired by the Nintendo Switch game 12Switch. The aim of the game is to strike a yoga pose and hold it while keeping the micro:bit completely still. If you wobble, your screen starts to fill up. If it is full, then you are out.

There are no extra components needed for this game as it just uses the built-in accelerometer on the x-axis to detect how much it wobbles rotationally left and right. You could tape the micro:bit and battery to a piece of card to make it easier to hold flat.

The code

We start by creating a variable called 'wobble' and setting it to 0 at the start. The value of 'wobble' is shown as a bar graph on the display. If 'wobble' becomes greater than 10, it shows an X and that player is out. If the accelerometer detects a tilt to the right or left exceeding 500 or -500, it adds 1 to your 'wobble' value.

Credit: Twitter user @stulowe80

Project scoring rubric

Assessment elements	1	2	3	4
Accelerometer	No working and/or meaningful use of accelerometer.	Use of accelerometer is incomplete or non-functional and/or tangential to operation of program.	Effectively uses the accelerometer with meaningful actions and responses.	Effectively uses the accelerometer with multiple meaningful actions and responses for each.
micro:bit program	micro:bit program lacks three or more of the required elements.	micro:bit program lacks two of the required elements.	micro:bit program lacks one of the required elements.	micro:bit program: - Effectively uses the Accelerometer with meaningful actions and responses - Compiles and runs as intended - Uses meaningful comments in code

Reflection Diary

Expectations

Write a reflection of about 150–300 words addressing the following points:

- What kind of project did you do? How did you decide what to pick?
- How does your project use the Accelerometer?
- Describe something in your project that you are proud of.
- Describe a difficult point in the process of designing this program and explain how you resolved it.
- What feedback did your testers give you? How did that help you improve your design?
- How would you improve your project given more time?
- Relating to the pair programming process (if applicable):
 - What challenges did you encounter working with a partner?
 - What benefits did you gain?
- Publish your MakeCode program and include the link.

Diary entry scoring rubric

Assessment elements	1	2	3	4
Diary entry	Diary entry is missing three or more of the required elements.	Diary entry is missing two of the required elements.	Diary entry is missing one of the required elements.	Diary entry addresses all elements.

Glossary of key terms

Acceleration: A measurement of an object's change in velocity over time. Acceleration is positive if an object is speeding up and negative if an object is slowing down.

G-force: A measurement of acceleration due to the Earth's gravity, typically measured in Gs where one G is equivalent to the mass of that object on earth.

Velocity: A measurement of an object's speed in terms of its rate of change in a specific direction.